Reinterpretation as an Operation, and the Boundary of Automated Physics Discovery

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A breakthrough in physics is rarely a better fit to the data nowadays. More often it is a decision that the data was about something else: Lorentz and Einstein predicted the same numbers, and Einstein's theory dropped an aether velocity no measurement could reach. Automated discovery has no operation for that move. Annotating 2,327 scientific prizes by what the contribution did, we find that among physics prizes **65% changed a description**, while the rung machine learning implements—fitting inside a fixed description—covers **42 of 542** depth labels, one in thirteen. Placing fourteen automated-discovery systems on the same ladder, all sit on that rung or one above it, and none performs the Lorentz-to-Einstein move. We give an operation that does. Structure no measurement can see is the null space of the prediction Jacobian, and removing it is reinterpretation. Applied naively this deletes conservation laws, because a global symmetry is invisible to every probe as well, so the operation must first ask whether the invariance survives being applied *point by point*. On a lattice gauge system, given only candidate generators and told nothing about which are real, the mechanism finds five invariant directions, separates one redundancy from four symmetries, and emits four conserved charges accurate to 10^{-10}; without the sort it would have destroyed all four. We then bound the result. On a real controversy—galaxy rotation curves—the criterion is silent, because it requires two descriptions that agree on the data. And four of eight Standard Model concepts were bought by a theorem before any experiment, which no probe-based method can reach. Data, code and negative results are released together at https://real-project-aether.github.io/project-aether/.

I. PHYSICS DISCOVERY IS MOSTLY REFRAMING

From the outside, science advances by measuring better and fitting better. From the inside, the decisive step is usually something else: someone changes what the measurements are taken to be about, and the old description loses a part it never needed.

In 1905 two theories fitted every optical and electrodynamic measurement equally well. Lorentz kept absolute space and added a contraction hypothesis and a local-time auxiliary, so that the aether became exactly undetectable [1]. Einstein removed the aether [2]. Nothing in the data chose between them: the choice was about which theory carried structure nothing could ever observe.

The distinction is old. Kuhn separated normal science from revolution [3], and Lakatos called an auxiliary that rescues a theory without predicting anything new *ad hoc* [4]. Both verdicts are delivered in retrospect and in prose. What is missing is one that can be computed while the outcome is still open—and an operation that performs the move rather than recognising it afterwards.

a. How much of physics is this? We annotated a corpus of scientific prizes by *what the contribution did* rather than by how it was cited. Existing prize datasets are bibliometric—who won, what they published, how it was cited [5]—which cannot say whether an advance reframed something or measured something new. A contribution changes one of three things: the **facts** you hold, the **instruments** you have, or the **description** you use. Only the third has a depth, and Fig. 1 gives that ladder. Its six rungs, in order, are **regression** (L0), solving inside the description you already have; **reinterpretation** (L1), the same predictions carried with less structure; **retyping** (L2), changing the level or the quantity the description is written in; **transfer** (L3), carrying one field's language into another; **new object** (L4), adding a kind of thing the language did not have; and **new move** (L5), changing what counts as a move at all. We use these names throughout, and on the project site.

The release holds **2,327 award records** across **23 prizes**, 1731–2026, of which **2,221** carry the awarding body's own citation and **1,547** are annotated. Restricting to the ten physics prizes—the physics Nobel, Dirac, Boltzmann, Onsager, Wolf, Heineman, Kavli, Sakurai, Lorentz and Breakthrough—gives 532 events, and the contrast with science at large is why this paper is about physics:

	all fields	**physics**
changed a description	45%	**65%**
changed no description at all	53%	**33%**

Across science most awarded work is not a change of description at all, and a framework built only around reframing would miss half its subject. In physics that inverts: two-thirds of awarded work changes the description, so the ladder is not a side issue but the main road. On it, **L0—work inside a fixed description—carries**

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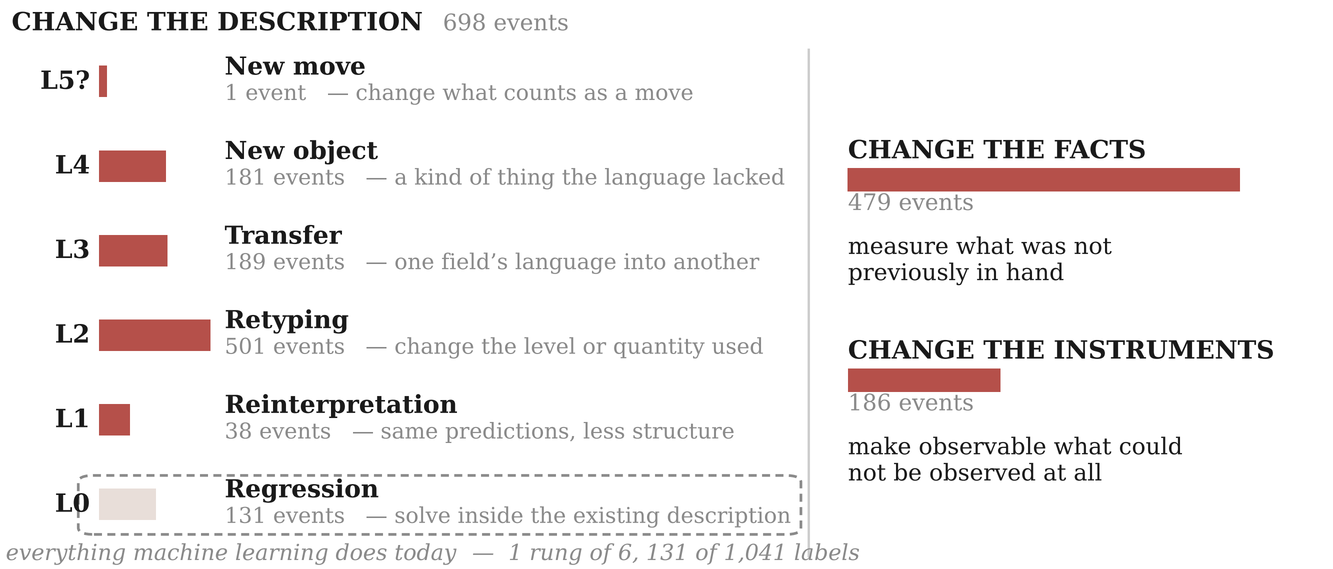

FIG. 1. What a discovering system has to be able to do. A move changes the facts you hold, the instruments you have, or the description you use; a change to the description has a depth. Counts are annotated prize events across all fields. Ordinary machine learning implements the bottom rung.

42 of the 542 physics depth labels, one in thirteen. L0 is not the shallow end; a Fields Medal can sit there. It is simply the only rung for which machine learning has an operation.

II. WHAT AUTOMATED DISCOVERY DOES TODAY

Table I places fourteen systems on the same ladder. We classify by the *operation the system exposes*, not by the significance of its results, and the claim is about published operations rather than about limits in principle.

The shape is consistent. Systems that search harder inside a language stay at L0 however impressive the result; DreamCoder reaches L2 by naming reusable structure; predicate invention aims at L4 and is still open. **Nothing performs L1.** No system takes a theory that already fits and returns a theory that fits identically while carrying less.

III. THE GAP, AND WHY A LANGUAGE MODEL DOES NOT CLOSE IT

A language model can recount the Lorentz–Einstein episode in detail, so it is fair to ask whether the operation is already available in a system that has read the histories. We put five episodes—three historical, two constructed controls—to a 20B-parameter open-weight model [20] in three conditions. The historical three are supernova distance moduli [21] for the acceleration reported in 1998–99 [22, 23] against the grey-dust patch offered for the same dimming [24]; relativistic mass against Lorentz's aether theory; and quasicrystal diffraction [25] against Pauling's multiple twinning [26]. The conditions are **named**, with real theories and domain words; **blind**, every identifier alpha-renamed but fit statistics retained; and **bare**, renamed with statistics withheld.

Table II gives the result. Accuracy is far above chance when the model can recognise the episode ($p = 5\times10^{-9}$) and *indistinguishable from chance* when it cannot ($p =$ 0.9 blind, 0.7 bare; named against blind, paired $p =$ 0.03). The model is recalling these episodes, not judging them—renaming alone is already known to move accuracy on grade-school arithmetic [27], and the blind condition is that probe applied to discovery. One case survives anonymisation: relativity stays at 100% because $(1 - v^2)^{-1/2}$ is recognisable from its algebra whatever the identifiers are called. Handing over the statistics does not rescue the others, so the failure is not a missing-information problem.

IV. AN OPERATION FOR REINTERPRETATION

If no system performs the move, the first question is whether it can be written down as something a machine runs at all. A verdict available only in hindsight and in prose cannot be automated, so what follows turns "Einstein's theory carried less than Lorentz's" into a quantity computable from a theory and its data before anyone knows which one is right.

a. The measurement. "Carries structure no measurement can see" is not a metaphor. For a theory with pa-

TABLE I. Automated discovery systems placed on the ladder of Fig. 1. Placement is by the operation each system exposes. Everything clusters on L0; library learning reaches L2; predicate invention aims at L4 and remains open. **No system implements L1.**

system	what it does	rung	why not higher
BACON [6]	rediscovers laws from tabulated data	L0	searches within a fixed term language
Eureqa [7]	symbolic regression on motion data	L0	the operator set is given in advance
AI Feynman [8]	physics-informed symbolic regression	L0	recovers known forms; invents no term
PySR [9]	evolutionary symbolic regression	L0	fixed primitive set
LaSR [10]	symbolic regression with a learned library	L0/L2	abstractions are shortcuts, not retypings
DreamCoder [11]	wake–sleep library learning	**L2**	names reusable structure; no theory to reinterpret
FunSearch [12]	LLM program search against an evaluator	L0	searches programs in a fixed language
AlphaGeometry [13]	olympiad geometry by synthesis and search	L0	derives inside a fixed axiom set
Robot Scientist [14]	forms and tests hypotheses with a robot	L0 + facts	chooses experiments; the ontology is fixed
Coscientist [15]	plans and runs chemistry autonomously	instruments	extends reach; no description changes
AI Scientist [16]	end-to-end machine-learning papers	L0	searches a fixed hypothesis space
AlphaFold [17]	protein structure prediction	tractability	makes computable; does not reframe
Cascade-Correlation [18]	grows hidden units as needed	L0	capacity, not concepts
predicate invention [19]	invents relations in logic programs	**L4***	*open after three decades of ILP

rameters p predicting $f(x, p)$, the directions along which no prediction moves are the null space of the Jacobian $J_{ij} = \partial f_i/\partial p_j$. This is structural non-identifiability [28], read as a property of a theory rather than a nuisance in a fit. Exact degeneracy is the wrong test for real data—Lorentz's aether was not unobservable in principle, but to within the precision of the day—so we scale column j by $|p_j|$ and row i by $1/\sigma_i$, and ask whether a direction can be moved by an order-unity amount without shifting any prediction past its error bar.

b. Why removing it is not enough. Quotienting every invisible direction is wrong, and wrong in a way that matters. A *global* phase rotation is invisible to every probe exactly as a gauge transformation is. Removing both does not discover charge conservation; it **erases** the conserved quantity. One further question separates them:

invariant *point by point*	redundancy: quotient it [L1]
invariant only *uniformly*	symmetry: keep it, read off a conserved charge [L4]

This turns one operation into two, and the second is the more productive: *a distinction no probe can see, but only when moved everywhere at once, is a conservation law.* Our implementation refuses to quotient at all when no locality test is available, rather than guess.

c. What proposes. A null-space search finds only what its basis contains, so something must widen the basis. Section III says what a language model is for here: it recalls fluently and judges at chance. We therefore use it *only to generate* and never to decide.

TABLE II. Accuracy at identifying which of two candidates is the reinterpretation. Twelve repeats per cell, presentation order randomised; the two controls are averaged. Chance is 50%.

episode	named	blind	bare
dark energy	67%	42%	50%
relativity	100%	100%	100%
quasicrystals	100%	50%	42%
two controls	84%	25%	38%
pooled, 60 trials	**87%**	48%	53%

V. RESULTS

a. Controls first. Twenty-one checks run on cases whose answers we put in ourselves: the mechanism must find hidden slack and remove it *without moving any prediction*, leave an honest theory alone, and **reject** a proposed term that buys coverage—the fraction of observations the fitted theory reproduces inside 2σ—while moving no prediction—the property that stops "add a parameter" from being free. All twenty-one pass. This establishes that the implementation does what it claims, not that the criterion tracks real theory choice; the two results below are where that is tested.

b. Discovering an unknown symmetry. On a periodic 6×6 lattice carrying a complex two-flavour matter field and U(1) links, probed only by gauge-invariant, flavour-blind quantities, the mechanism is handed eight candidate generators and told nothing about which are real invariances. The null space of $\partial(\text{probes})/\partial c$ under uniform application has rank 3, so **five invariant directions are found**; a link phase, a rescale and a single-flavour shift are rejected. A second null-space computation *inside* the first, with the coefficient now varying from site to site, isolates **one redundancy** ($+1.00$ gauge, local response 2.6×10^{-5}) from **four symmetries** (local response 5.1–6.6×10^{1}). The separation had to be done in that order: at uniform order the gauge transformation and the global phase act identically on the state, so the first null space mixes them and only the local test breaks the degeneracy.

Every symmetry branch then pays, and the redundancy branch does not:

branch	relative drift
four symmetry charges	1.8×10^{-11}–7.6×10^{-10}
redundancy, would-be charge	8.6×10^{-1}
energy, same trajectory	1.3×10^{-10}

The charges are conserved at the trajectory's own precision and the redundancy's would-be charge is nine orders of magnitude worse, which is the evidence that the sort is real rather than a relabelling. **Without the local test, all five directions are unobservable and all five would be quotiented—destroying four conservation laws in order to remove one redundancy.**

c. The proposer, and its filter. On symbolic worlds the model proposes candidate concepts and the acceptance rule decides. In a world whose residual hides $0.4\sin 3x$ it proposed eight concepts and two to three were accepted; in a control whose residual is pure noise it proposed eight and **none** was accepted. The model proposes with equal fluency in both; only the measurement tells them apart. Offered the true concept directly the filter is decisive—$\sin 3x$ reaches 97% coverage against 68% for $\sin 2.9x$ and 20% for wrong terms—so the limiting component here is the proposer, not the test: across runs the model offered $\sin x$, $\sin 2x$, x^3 and $\sin \pi x$, never $\sin 3x$, though the residual is a clean sine.

d. Where the criterion goes silent. We applied it to a live controversy with no known answer: the rotation curve of NGC 3198 from SPARC [29], fitted by a dark-matter halo and by MOND [30].

description	concepts	χ^2/N	unobservable
baryons + NFW halo	3	1.22	**0**
MOND, a_0 universal	1	9.21	**0**
MOND, a_0 free	2	1.78	**0**

Neither description carries structure the measurements cannot see: the halo model's softest direction, a mass-to-light against concentration trade-off, still moves the curve 6.4σ under a 100% excursion—unsurprising, since SPARC's $3.6\,\mu$m photometry was assembled to break exactly that degeneracy. **The criterion is silent, and that is the result.** The Lorentz–Einstein signature needs two descriptions that *agree* on the data and differ in what they carry; these disagree at χ^2/N of 1.22 against 9.21, and where descriptions disagree ordinary model comparison settles the matter. The operation applies to a narrower class of disputes than we assumed. Our free-a_0 fit gives 0.67×10^{-8} cm s^{-2} against the canonical 1.22×10^{-8}; NGC 3198 is a documented borderline case preferring a lower a_0, so we reproduce the known tension rather than contradicting it.

VI. WHAT PROBE-BASED DISCOVERY CANNOT REACH

The mechanism above buys concepts with measurements. Tested against eight concept-forming episodes in the Standard Model it accounts for **two**: the neutrino, forced by a continuous spectrum where a two-body decay demands a line, and gauge redundancy. The other six do not fail for want of cleverness. Four of them contain *no probe at all*:

- Δ^{++} violated Fermi statistics. Colour was invented so the state would be *allowed*.
- Flavour-changing neutral currents were too suppressed. Charm was invented so the excess would *cancel*.
- Gauge anomalies had to sum to zero, which required *complete* generations.
- WW scattering had to stay unitary, which put the Higgs below about a TeV—and told experimenters what to build.

Every one predicted a real particle. Every one was paid for by a **theorem first**, with the experiment arriving years or decades later. A framework whose founding commitment is that a concept is paid for by an experiment has these exactly backwards. The prerequisite is not more data or a better search: *you can only have theorems if there is a formal theory to have them about*, and no discovery system we know of carries one. That, rather than the difficulty of reinterpretation, is what bounds automated physics discovery today.

The weaker version of this route is already refuted here. Proposing concepts from structural defects in a description—dangling structure, role collisions—was tested on three substrates and lost on all three. The difference between that and the Standard Model is grade, not kind: heuristics say a defect is a hint, theorems say a state is forbidden or it is not.

VII. LIMITATIONS

a. One annotator, and it is a model. Every count in Fig. 1 and in Sec. II is one annotator's judgement, and that annotator is a large language model reading each official citation under human direction. These are not the judgements of human domain experts; there is exactly one annotator, so no inter-annotator agreement statistic exists and we do not report a proxy for one. The boundary that matters most is between recording facts and changing a description, and it is a judgement call. Independent reannotation is the single most useful contribution a reader could make.

b. The criterion was revised against known answers. An earlier form of this test was modified eight times, each after seeing a verdict we knew to be wrong; a description-length score [31] was among the versions that failed. Five episodes whose outcomes we already knew establish internal consistency, not predictive power, which is why the results above rest instead on a lattice system with an independently known answer and on a controversy with no known answer at all.

c. The acceptance rule does not discriminate. "Coverage strictly improves" admitted $\sin x$ and x^3 at 20–22% against a base of 18%. It ranks correctly but needs a margin, not a strict inequality.

d. The scheme is not symmetric. The facts column distinguishes facts newly in hand from reach newly extended, but encodes that as two categories rather than as a ladder. A scheme that graded every column would be cleaner; we did not build one.

e. L2 is a residual category, not a move. Of its 501 events, 34% carry a coarse-graining operator and only 7% carry the merge-or-split operator by which we first defined the rung; a further 10% and 7% carry operators belonging to L4 and L3. L2 is where events land when they change the description but are not cleanly L1, L3 or L4. We have renamed it after its dominant sub-move, changing the level or quantity a description is written in, but the honest reading is that this rung should be cut into two or three.

f. Hindsight and selection. Prizes are awarded with decades of hindsight, so the corpus samples work that turned out to matter, and citations justify a decision already taken.

g. Availability. Everything is at `https://real-project-aether.github.io/project-aether/` — the corpus, the mechanism, both negative results, and a script that recomputes every figure in this paper from the shipped data and fails loudly if one does not reproduce.

[1] H. A. Lorentz, Proc. R. Neth. Acad. Arts Sci. **6**, 809 (1904), reprinted in Collected Papers Vol. 5 (1937).

[2] A. Einstein, Annalen der Physik **322**, 891 (1905).

[3] T. S. Kuhn, *The Structure of Scientific Revolutions* (University of Chicago Press, Chicago, 1962).

[4] I. Lakatos, in *Criticism and the Growth of Knowledge*, edited by I. Lakatos and A. Musgrave (Cambridge University Press, Cambridge, 1970) pp. 91–196.

[5] J. Li, Y. Yin, S. Fortunato, and D. Wang, Scientific Data **6**, 33 (2019).

[6] P. Langley, Cognitive Science **5**, 31 (1981).

[7] M. Schmidt and H. Lipson, Science **324**, 81 (2009).

[8] S.-M. Udrescu and M. Tegmark, Science Advances **6**, eaay2631 (2020).

[9] M. Cranmer, arXiv:2305.01582 (2023).

[10] A. Grayeli, A. Sehgal, O. Costilla-Reyes, M. Cranmer, and S. Chaudhuri, in *Advances in Neural Information Processing Systems 37* (Neural Information Processing Systems Foundation, Inc. (NeurIPS), Vancouver, BC, Canada, 2024) pp. 44678–44709, arXiv:2409.09359.

[11] K. Ellis, C. Wong, M. Nye, M. Sablé-Meyer, L. Morales, L. Hewitt, L. Cary, A. Solar-Lezama, and J. B. Tenenbaum, in *Proc. 42nd ACM SIGPLAN International Conference on Programming Language Design and Implementation (PLDI)* (ACM, Virtual Canada, 2021) pp. 835–850.

[12] B. Romera-Paredes, M. Barekatain, A. Novikov, M. Balog, M. P. Kumar, E. Dupont, F. J. R. Ruiz, J. S. Ellenberg, P. Wang, O. Fawzi, *et al.*, Nature **625**, 468 (2024).

[13] T. H. Trinh, Y. Wu, Q. V. Le, H. He, and T. Luong, Nature **625**, 476 (2024).

[14] R. D. King, K. E. Whelan, F. M. Jones, P. G. K. Reiser, C. H. Bryant, S. H. Muggleton, D. B. Kell, and S. G. Oliver, Nature **427**, 247 (2004).

[15] D. A. Boiko, R. MacKnight, B. Kline, and G. Gomes, Nature **624**, 570 (2023).

[16] C. Lu, C. Lu, R. T. Lange, J. Foerster, J. Clune, and D. Ha, arXiv:2408.06292 (2024).

[17] J. Jumper, R. Evans, A. Pritzel, T. Green, *et al.*, Nature **596**, 583 (2021).

[18] S. E. Fahlman and C. Lebiere, in *Advances in Neural Information Processing Systems 2* (1990) pp. 524–532.

[19] A. Cropper and S. Dumančić, Journal of Artificial Intelligence Research **74**, 765 (2022).

[20] OpenAI, gpt-oss-120b and gpt-oss-20b model card (2025), arXiv:2508.10925.

[21] D. M. Scolnic *et al.*, Astrophysical Journal **859**, 101 (2018).

[22] A. G. Riess *et al.*, Astronomical Journal **116**, 1009 (1998).

[23] S. Perlmutter *et al.*, Astrophysical Journal **517**, 565 (1999).

[24] A. N. Aguirre, Astrophysical Journal Letters **512**, L19 (1999).

[25] D. Shechtman, I. Blech, D. Gratias, and J. W. Cahn, Physical Review Letters **53**, 1951 (1984).

[26] L. Pauling, Nature **317**, 512 (1985).

[27] I. Mirzadeh, K. Alizadeh, H. Shahrokhi, O. Tuzel, S. Bengio, and M. Farajtabar, in *International Conference on Learning Representations (ICLR)* (2025).

[28] A. Raue, C. Kreutz, T. Maiwald, J. Bachmann, M. Schilling, U. Klingmüller, and J. Timmer, Bioinformatics **25**, 1923 (2009).

[29] F. Lelli, S. S. McGaugh, and J. M. Schombert, Astron. J. **152**, 157 (2016).

[30] M. Milgrom, Astrophys. J. **270**, 365 (1983).

[31] J. Rissanen, Automatica **14**, 465 (1978).